Problem

The current I in the circuit of Fig. 2.63 is:

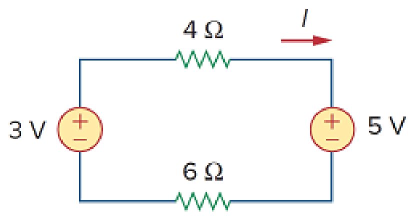
(a) −0.8 A

(b) −0.2 A

(c) 0.2 A

(d) 0.8 A

Figure 2.63:



Step-by-step solution

Step 1 of 2

Refer to the circuit in Figure 2.63 in the textbook.

Redraw the circuit as shown in Figure 1.

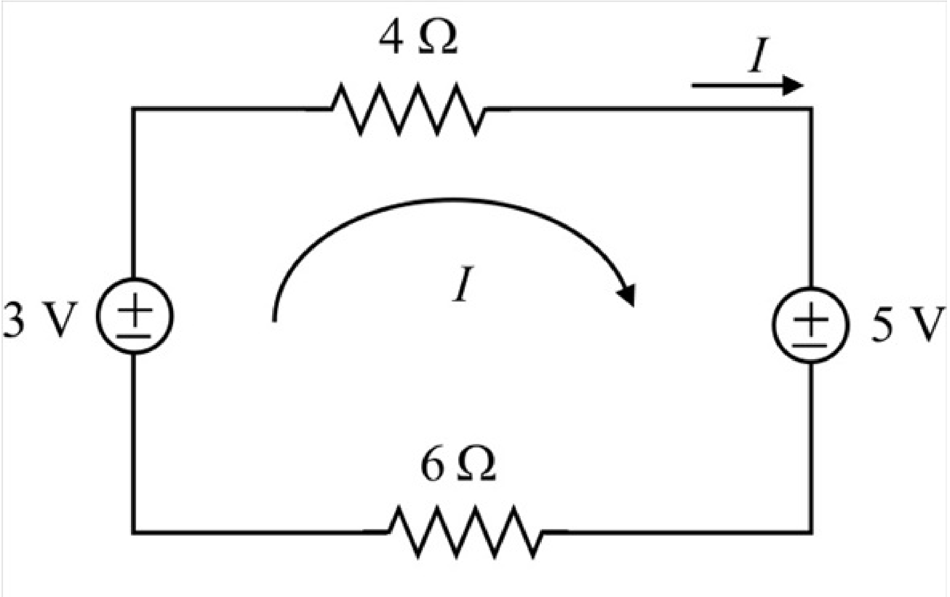


Figure 1

Step 2 of 2

Apply Kirchhoff's voltage law to the circuit in Figure 1.

$$-3+4I+5+6I=0$$

$$10I+2=0$$

$$10I=-2$$

$$\begin{aligned} I &= \frac{-2}{10} \\ &= -0.2\ \text{A} \end{aligned}$$

Thus, the value of current *I* is $\boxed{-0.2\ \text{A}}$.